

Chapter 10

Direct and Inverse Variations

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Exercise 10(A)

Question 1(i)

The values of a and b from the table are:

x	y
3	18
5	a



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x	y
b	90
8	48

1. $a = 30$ and $b = 270$

2. $a = 30$ and $b = 15$

3. $a = 45$ and $b = 15$

4. $a = 30$ and $b = 540$

Answer

$$\frac{3}{18} = \frac{1}{6}$$

$$\frac{8}{48} = \frac{1}{6}$$

So, we can say that x and y vary directly

$$\therefore \frac{3}{18} = \frac{5}{a} = \frac{b}{90} = \frac{8}{48}$$

$$\Rightarrow \frac{3}{18} = \frac{5}{a} \text{ and } \frac{3}{18} = \frac{b}{90}$$

$$\Rightarrow a = \frac{5 \times 18}{3} \text{ and } b = \frac{3 \times 90}{3}$$

$$\Rightarrow a = \frac{90}{3} \text{ and } b = \frac{270}{3}$$

$$\Rightarrow a = 30 \text{ and } b = 90$$

Hence, option 2 is the correct option.

Question 1(ii)

In 5 hours, a machine makes 45 screws. In 12 hours, the same machine will make:

1. 36 screws
2. 90 screws
3. 72 screws
4. 108 screws

Answer

Let the machine make x screws.

Hours	Screws
5	45
12	x

Since it is a case of direct variation.

$$\therefore \frac{5}{45} = \frac{12}{x}$$

$$\Rightarrow x = \frac{45 \times 12}{5}$$

$$\Rightarrow x = \frac{540}{5}$$

$$\Rightarrow x = 108$$

Hence, option 4 is the correct option.

Question 1(iii)

If the cost of 9 pens is ₹ 369, the cost of one dozen pens of the same make will be:

1. ₹ 42
2. ₹ 429
3. ₹ 942
4. ₹ 492

Answer

Let the cost of one dozen pens be x.

Pens	Cost
9	369
12	x

Since it is a case of direct variation.

$$\therefore \frac{9}{369} = \frac{12}{x}$$

$$\Rightarrow x = \frac{369 \times 12}{9}$$

$$\Rightarrow x = \frac{4,428}{9}$$

$$\Rightarrow x = 492$$

Hence, option 4 is the correct option.

Question 1(iv)

If y varies directly to x and y = 80 when x = 400; the value of y when x = 25 is:

1. 5
2. 25
3. 75
4. 125

Answer

Let the value of y when x = 25 be a

x	y
400	80
25	a

Since it is a case of direct variation,

$$\therefore \frac{400}{80} = \frac{25}{a}$$

$$\Rightarrow a = \frac{80 \times 25}{400}$$

$$\Rightarrow a = \frac{2,000}{400}$$

$$\Rightarrow a = 5$$

Hence, option 1 is the correct option.

Question 1(v)

A man working 48 hours per week earns ₹ 2,400. If he works for 36 hours per week, he will earn:

1. ₹ 3,200

2. ₹ 900

3. ₹ 1,800

4. ₹ 1,080

Answer

Let the man earn ₹ x.

Hours	Earning
48	₹ 2,400
36	₹ x

Since it is a case of direct variation.

$$\therefore \frac{48}{2400} = \frac{36}{x}$$

$$\Rightarrow x = \frac{36 \times 2400}{48}$$

$$\Rightarrow x = \frac{86,400}{48}$$

$$\Rightarrow x = 1,800$$

Hence, option 3 is the correct option.

Question 2(i)

In the following table, do x and y vary directly:

x	y
3	4.5
5	7.5
8	12
11	16.5

Answer

$$\frac{x_1}{y_1} = \frac{3}{4.5} = \frac{2}{3},$$

$$\frac{x_2}{y_2} = \frac{5}{7.5} = \frac{2}{3},$$

$$\frac{x_3}{y_3} = \frac{8}{12} = \frac{2}{3},$$

$$\frac{x_4}{y_4} = \frac{11}{16.5} = \frac{2}{3},$$

$$\Rightarrow \frac{x_1}{y_1} = \frac{x_2}{y_2} = \frac{x_3}{y_3} = \frac{x_4}{y_4} = \text{a constant}$$

Hence, x and y are in direct variation.

Question 2(ii)

In the following table, do x and y vary directly:

x	y
16	32
30	60
40	80
56	84

Answer

$$\frac{x_1}{y_1} = \frac{16}{32} = \frac{1}{2},$$

$$\frac{x_2}{y_2} = \frac{30}{60} = \frac{1}{2},$$

$$\frac{x_3}{y_3} = \frac{40}{80} = \frac{1}{2},$$

$$\frac{x_4}{y_4} = \frac{56}{84} = \frac{2}{3},$$

$$\Rightarrow \frac{x_1}{y_1} = \frac{x_2}{y_2} = \frac{x_3}{y_3} \neq \frac{x_4}{y_4} = \text{not a constant}$$

Hence, x and y are not in direct variation.

Question 2(iii)

In the following table, do x and y vary directly:

x	y
27	81
45	180
54	216
75	225

Answer

$$\frac{x_1}{y_1} = \frac{27}{81} = \frac{1}{3},$$

$$\frac{x_2}{y_2} = \frac{45}{180} = \frac{1}{4},$$

$$\frac{x_3}{y_3} = \frac{54}{216} = \frac{1}{4},$$

$$\frac{x_4}{y_4} = \frac{75}{225} = \frac{1}{3},$$

$$\Rightarrow \frac{x_1}{y_1} \neq \frac{x_2}{y_2} = \frac{x_3}{y_3} \neq \frac{x_4}{y_4} = \text{not a constant}$$

Hence, x and y are not in direct variation.

Question 3

If x and y vary directly, find the values of x, y and z :

x	y
3	36
x	60
y	96
10	z

Answer

Since a and b vary directly,

$$\frac{3}{36} = \frac{x}{60} = \frac{y}{96} = \frac{10}{z}$$

$$\Rightarrow \frac{3}{36} = \frac{x}{60}, \frac{3}{36} = \frac{y}{96} \text{ and } \frac{3}{36} = \frac{10}{z}$$

$$\Rightarrow x = \frac{3 \times 60}{36}, y = \frac{3 \times 96}{36} \text{ and } z = \frac{36 \times 10}{3}$$

$$\Rightarrow x = \frac{180}{36}, y = \frac{288}{36} \text{ and } z = \frac{360}{3}$$

$$\Rightarrow x = 5, y = 8 \text{ and } z = 120$$

Hence, $x = 5$, $y = 8$ and $z = 120$.

Question 4

A truck consumes 28 litres of diesel for moving through a distance of 448 km.

How much distance will it cover in 64 litres of diesel?

Answer

Let the truck consume x litres of diesel.

Diesel	Distance
28	448
64	x

Since it is a case of direct variation.

$$\therefore \frac{28}{448} = \frac{64}{x}$$

$$\Rightarrow x = \frac{64 \times 448}{28}$$

$$\Rightarrow x = \frac{28,672}{28}$$

$$\Rightarrow x = 1,024$$

Hence, the truck will cover 1,024 km distance.

Question 5

For 100 km, a taxi charges ₹ 1,800. How much will it charge for a journey of 120 km ?

Answer

Let the taxi charges ₹ x for 120 km.

Distance	Charge
100	₹ 1,800
120	₹ x

Since it is a case of direct variation,

$$\therefore \frac{100}{1,800} = \frac{120}{x}$$

$$\Rightarrow x = \frac{120 \times 1800}{100}$$

$$\Rightarrow x = \frac{2,16,000}{100}$$

$$\Rightarrow x = 2,160$$

Hence, the taxi will charge ₹ 2,160.

Question 6

If 27 identical articles cost ₹ 1,890, how many articles can be bought for ₹ 1,750 ?

Answer

Let x articles cost ₹ 1,750.

Articles	Cost
27	₹ 1,890
x	₹ 1,750

Since it is a case of direct variation.

$$\therefore \frac{27}{1,890} = \frac{x}{1,750}$$

$$\Rightarrow x = \frac{27 \times 1,750}{1,890}$$

$$\Rightarrow x = \frac{47,250}{1,890}$$

$$\Rightarrow x = 25$$

Hence, the 25 articles can be bought for ₹ 1,750.

Question 7

7 kg of rice costs ₹ 1,120. How much rice can be bought for ₹ 3,680?

Answer

Let x kg rice cost ₹ 3,680.

Quantity	Cost
7	₹ 1,120

Quantity	Cost
x	₹ 3,680

Since it is a case of direct variation,

$$\therefore \frac{7}{1,120} = \frac{x}{3,680}$$

$$\Rightarrow x = \frac{7 \times 3,680}{1,120}$$

$$\Rightarrow x = \frac{25,760}{1,120}$$

$$\Rightarrow x = 23$$

Hence, the 23 kg of rice can be bought for ₹ 3,680.

Question 8

6 notebooks cost ₹ 156, find the cost of 54 such notebooks.

Answer

Let 54 notebooks cost ₹ x.

Quantity	Cost
6	₹ 156
54	₹ x

Since it is a case of direct variation,

$$\begin{aligned}\therefore \frac{6}{156} &= \frac{54}{x} \\ \Rightarrow x &= \frac{54 \times 156}{6} \\ \Rightarrow x &= \frac{8,424}{6} \\ \Rightarrow x &= 1,404\end{aligned}$$

Hence, the cost of 54 notebooks is ₹ 1,404.

Question 9

22 men can dig a 27 m long trench in one day. How many men should be employed for digging 135 m long trench of the same type in one day ?

Answer

Let the x men be employed for digging 135 m long trench in one day.

Men	Length
22	27
x	135

Since it is a case of direct variation,

$$\begin{aligned}\therefore \frac{22}{27} &= \frac{x}{135} \\ \Rightarrow x &= \frac{22 \times 135}{27}\end{aligned}$$

$$\Rightarrow x = \frac{2,970}{27}$$

$$\Rightarrow x = 110$$

Hence, 110 men should be employed for digging 135 m long trench in one day.

Question 10

If the total weight of 11 identical articles is 77kg, how many articles of the same type would weigh 224 kg?

Answer

Let x articles weigh 224 kg.

Articles	Weight
11	77
x	224

Since it is a case of direct variation,

$$\therefore \frac{11}{77} = \frac{x}{224}$$

$$\Rightarrow x = \frac{11 \times 224}{77}$$

$$\Rightarrow x = \frac{2,464}{77}$$

$$\Rightarrow x = 32$$

Hence, 32 articles would weigh 224 kg.

Question 11

A train is moving with uniform speed of 120 km per hour.

(i) How far will it travel in 36 minutes?

(ii) In how much time will it cover 210 km?

Answer

(i) Let x km be distance travel in 36 minutes.

Distance	Time (in minutes)
120	60 ($\because$ 1 hour = 60 mins)
x	36

Since it is a case of direct variation,

$$\therefore \frac{120}{60} = \frac{x}{36}$$

$$\Rightarrow x = \frac{120 \times 36}{60}$$

$$\Rightarrow x = \frac{4,320}{60}$$

$$\Rightarrow x = 72$$

Hence, the train will travel a distance of 72 km in 36 minutes.

(ii) Let the train take y minutes to cover 210 km.

Distance	Time (in minutes)
120	60
210	y

Since it is a case of direct variation,

$$\therefore \frac{120}{60} = \frac{210}{y}$$

$$\Rightarrow y = \frac{210 \times 60}{120}$$

$$\Rightarrow y = \frac{12,600}{120}$$

$$\Rightarrow y = 105$$

105 minutes = 60 minutes + 45 minutes

= 1 hours 45 minutes

Hence, the train will take 1 hours 45 minutes to cover 210 km.

Exercise 10(B)

Question 1(i)

If a varies inversely to b and $b = 12$ when a is 8; the value of b when $a = 6$ is:

1. 72

2. 32

3. 12

4. 16

Answer

$$a_1 b_1 = a_2 b_2 \quad [\because a \text{ varies inversely to } b]$$

$$\Rightarrow 8 \times 12 = 6 \times b$$

$$\Rightarrow 96 = 6 \times b$$

$$\Rightarrow b = \frac{96}{6}$$

$$\Rightarrow b = 16$$

Hence, option 4 is the correct option.

Question 1(ii)

12 men can make a certain number of screws in 30 days. The number of screws, of the same type will be made by 24 men in:

1. 15 days
2. 30 days
3. 60 days
4. none of these

Answer

Let 24 men take x days to make the same number of screws.

Men	Days
12	30
24	x

Since it is a case of inverse variation,

$$\therefore 12 \times 30 = 24 \times x$$

$$\Rightarrow 360 = 24x$$

$$\Rightarrow x = \frac{360}{24}$$

$$\Rightarrow x = 15$$

Hence, option 1 is the correct option.

Question 1(iii)

A school has 8 periods in a day, each of 45 minutes duration. Assuming that the number of school hours to be the same and the school has 9 periods in a day, the duration of each period is:

1. 72 minute
2. $180/5$ minute
3. 40 minute
4. 60 minute

Answer

Let the duration of each period be x minutes.

Periods	Duration(in min)
8	45
9	x

Since it is a case of inverse variation,

$$\therefore 8 \times 45 = 9 \times x$$

$$\Rightarrow 360 = 9x$$

$$\Rightarrow x = \frac{360}{9}$$

$$\Rightarrow x = 40$$

Hence, option 3 is the correct option.

Question 1(iv)

From the given table find the values of a and b:

x	y
12	5
4	a
10	6
30	b

1. a = 8 and b = 25
2. a = 15 and b = 12
3. a = 12 and b = 90
4. a = 15 and b = 2

Answer

$$\therefore x_1 y_1 = 12 \times 5 = 60$$

$$x_3 y_3 = 10 \times 6 = 60$$

So, we can say that this is inverse variation.

$$\Rightarrow a_1 b_1 = a_2 b_2 \quad [\because a \text{ varies inversely to } b]$$

$$\Rightarrow 12 \times 5 = 4 \times a$$

$$\Rightarrow 60 = 4 \times a$$

$$\Rightarrow a = \frac{60}{4}$$

$$\Rightarrow a = 15$$

And, similarly

$$a_3 b_3 = a_3 b_3 \quad [\because a \text{ varies inversely to } b]$$

$$\Rightarrow 10 \times 6 = 30 \times b$$

$$\Rightarrow 60 = 30 \times b$$

$$\Rightarrow b = \frac{60}{30}$$

$$\Rightarrow b = 2$$

Hence, option 4 is the correct option.

Question 1(v)

A hostel has provisions for 200 students for 30 days. If 100 new students join the hostel, the same provisions will last for:

1. 45 days
2. 30 days
3. 60 days
4. 20 days

Answer

Original no. of students in hostel = 200

New no. of students in hostel = $200 + 100 = 300$

For 300 students, let the provisions last for x days.

Students	Provision
200	30
300	x

Since it is a case of inverse variation,

$$\therefore 200 \times 30 = 300 \times x$$

$$\Rightarrow 6000 = 300x$$

$$\Rightarrow x = \frac{6000}{300}$$

$$\Rightarrow x = 20$$

Hence, option 4 is the correct option.

Question 2(i)

Check whether x and y vary inversely or not

x	y
4	6
3	8
12	2
1	24

Answer

$$\therefore x_1y_1 = 4 \times 6 = 24,$$

$$x_2y_2 = 3 \times 8 = 24,$$

$$x_3y_3 = 12 \times 2 = 24,$$

$$x_4y_4 = 1 \times 24 = 24,$$

$$\text{So, } x_1y_1 = x_2y_2 = x_3y_3 = x_4y_4$$

Hence, x and y vary inversely.

Question 2(ii)

Check whether x and y vary inversely or not

x	y
30	60
120	30
60	30
24	75

Answer

$$\therefore x_1y_1 = 30 \times 60 = 1,800,$$

$$x_2y_2 = 120 \times 30 = 3,600,$$

$$x_3y_3 = 60 \times 30 = 1,800,$$

$$x_4y_4 = 24 \times 75 = 1,800,$$

$$\text{So, } x_1y_1 \neq x_2y_2 \neq x_3y_3 = x_4y_4$$

Hence, x and y do not vary inversely.

Question 2(iii)

Check whether x and y vary inversely or not

x	y
10	90
30	30
60	20

x	y
10	90

Answer

$$\therefore x_1y_1 = 10 \times 90 = 900,$$

$$x_2y_2 = 30 \times 30 = 900,$$

$$x_3y_3 = 60 \times 20 = 1,200,$$

$$x_4y_4 = 10 \times 90 = 900,$$

$$\text{So, } x_1y_1 = x_2y_2 \neq x_3y_3 \neq x_4y_4$$

Hence, x and y do not vary inversely.

Question 3(i)

If x and y vary inversely, find the values of l, m and n :

x	y
4	4
8	l
2	m
32	n

Answer

Since it is a case of inverse variation,

$$\therefore 4 \times 4 = 8 \times l = 2 \times m = 32 \times n$$

$$\Rightarrow 16 = 8l = 2m = 32n$$

$$\Rightarrow l = \frac{16}{8}, m = \frac{16}{2} \text{ and } n = \frac{16}{32}$$

$$\Rightarrow l = 2, m = 8 \text{ and } n = 0.5$$

Hence, l = 2, m = 8 and n = 0.5.

Question 3(ii)

If x and y vary inversely, find the values of l, m and n :

x	y
24	l
32	12
m	8
16	n

Answer

Since it is a case of inverse variation,

$$\therefore 24 \times l = 32 \times 12 = m \times 8 = 16 \times n$$

$$\Rightarrow 24l = 384 = 8m = 16n$$

$$\Rightarrow l = \frac{384}{24}, m = \frac{384}{8} \text{ and } n = \frac{384}{16}$$

$$\Rightarrow l = 16, m = 48 \text{ and } n = 24$$

Hence, l = 16, m = 48 and n = 24.

Question 4

36 men can do a piece of work in 7 days. How many men will do the same work in 42 days?

Answer

Let x men do the same work in 42 days.

Men	Days
36	7
x	42

Since it is a case of inverse variation.

$$\therefore 36 \times 7 = x \times 42$$

$$\Rightarrow 252 = 42x$$

$$\Rightarrow x = \frac{252}{42}$$

$$\Rightarrow x = 6$$

Hence, 6 men will do the same work in 42 days.

Question 5

12 pipes, all of the same size, fill a tank in 42 minutes. How long will it take to fill the same tank, if 21 pipes of the same size are used?

Answer

Let 21 pipes take x minutes to fill the tank.

Pipes	Time (in min)
12	42
21	x

Since it is a case of inverse variation,

$$\therefore 12 \times 42 = 21 \times x$$

$$\Rightarrow 504 = 21x$$

$$\Rightarrow x = \frac{504}{21}$$

$$\Rightarrow x = 24$$

Hence, 21 pipes will take 24 minutes to fill the tank.

Question 6

In a fort, 150 men had provisions for 45 days. After 10 days, 25 men left the fort. How long would the food last at the same rate?

Answer

Original no. of men in the fort = 150

New no. of men in the fort = $150 - 25 = 125$

For 125 men lets take food will last for x days.

The remaining food would last $(45 - 10) = 35$ days

Men	Provision
150	35
125	x

Since it is a case of inverse variation,

$$\therefore 150 \times 35 = 125 \times x$$

$$\Rightarrow 5,250 = 125x$$

$$\Rightarrow x = \frac{5,250}{125}$$

$$\Rightarrow x = 42$$

Hence, the food last for 42 days.

Question 7

72 men do a piece of work in 25 days. In how many days will 30 men do the same work?

Answer

Let 30 men do the same work in x days.

Men	Days
72	25
30	x

Since it is a case of inverse variation,

$$\therefore 72 \times 25 = 30 \times x$$

$$\Rightarrow 1,800 = 30x$$

$$\Rightarrow x = \frac{1,800}{30}$$

$$\Rightarrow x = 60$$

Hence, 30 men will do the same work in 60 days.

Question 8

If 56 workers can build a wall in 180 hours, how many workers will be required to do the same work in 70 hours?

Answer

Let x workers do the same work in 70 hours.

Workers	Time (in hours)
56	180
x	70

Since it is a case of inverse variation,

$$\therefore 56 \times 180 = x \times 70$$

$$\Rightarrow 10,080 = 70x$$

$$\Rightarrow x = \frac{10,080}{70}$$

$$\Rightarrow x = 144$$

Hence, 144 workers will do the work in 70 hours.

Question 9

A car takes 6 hours to reach a destination by travelling at the speed of 50 km per hour. How long will it take when the car travels at the speed of 75 km per hour?

Answer

Let the car take x hours at the speed of 75 km per hour.

Time (in hr)	Speed
6	50
x	75

Since it is a case of inverse variation,

$$\therefore 6 \times 50 = x \times 75$$

$$\Rightarrow 300 = 75x$$

$$\Rightarrow x = \frac{300}{75}$$

$$\Rightarrow x = 4$$

Hence, the car takes 4 hours at the speed of 75 km per hour.

Exercise 10(C)

Question 1(i)

A can do a piece of work in 2 days and B can do the same work in 3 days. If A and B work together, the amount of work done by them in 1 day is:

1. 30

2. 1.5

3. $\frac{2}{3}$

4. $\frac{5}{6}$

Answer

$$\text{A's 1 day work} = \frac{1}{2}$$

$$\text{B's 1 day work} = \frac{1}{3}$$

$$(\text{A} + \text{B})\text{'s 1 day work} = \frac{1}{2} + \frac{1}{3}$$

$$= \frac{3 + 2}{6}$$

$$= \frac{5}{6}$$

Hence, option 4 is the correct option.

Question 1(ii)

If $\frac{1}{20}$ of a work can be done in 5 days, the amount of work done in one day will be :

1. $\frac{1}{100}$

2. 100

3. 4

4. 5

Answer

Let work done in one day be x

$$\text{Work done in 5 days} = \frac{1}{20}$$

$$\therefore 5x = \frac{1}{20}$$

$$\Rightarrow x = \frac{1}{20 \times 5}$$

$$\Rightarrow x = \frac{1}{100}$$

Hence, option 1 is the correct option.

Question 1(iii)

Ritu can knit sweater in 4 days and Manish can knit the same sweater in 6 days.

If they together knit the same sweater, the number of days taken by them will be:

1. 10 days

2. 5 days

3. $\frac{12}{5}$ days

4. $\frac{5}{12}$ days

Answer

$$\text{Ritu's 1 day work} = \frac{1}{4}$$

$$\text{Manish's 1 day work} = \frac{1}{6}$$

$$\begin{aligned}(\text{Ritu} + \text{Manish})\text{'s 1 day work} &= \frac{1}{4} + \frac{1}{6} \\&= \frac{3 + 2}{12} \\&= \frac{5}{12}\end{aligned}$$

$$\begin{aligned}\text{No. days taken by Ritu \& Manish together to knit the same sweater} &= \frac{1}{\frac{5}{12}} = \\&= \frac{12}{5}\end{aligned}$$

Hence, option 3 is the correct option.

Question 1(iv)

A and B working together can complete a work in 4 days. If A alone can do the same work in 6 days, then B alone can do the same work in:

1. 12 days
2. 10 days
3. 2 days
4. 24 days

Answer

$$\begin{aligned}(\text{A} + \text{B})\text{'s 1 day work} &= \frac{1}{4} \\ \text{A's 1 day work} &= \frac{1}{6}\end{aligned}$$

$$\begin{aligned}\text{B's 1 day work} &= \frac{1}{4} - \frac{1}{6} \\ &= \frac{3 - 2}{12} \\ &= \frac{1}{12}\end{aligned}$$

B alone can do the work in 12 days

Hence, option 1 is the correct option.

Question 2

A can do a piece of work in 10 days and B in 15 days. How long will they take to finish it working together?

Answer

$$\begin{aligned}\text{A's 1 day work} &= \frac{1}{10} \\ \text{B's 1 day work} &= \frac{1}{15} \\ (\text{A} + \text{B})\text{'s 1 day work} &= \frac{1}{10} + \frac{1}{15} \\ &= \frac{3 + 2}{30} \\ &= \frac{5}{30} \\ \text{A and B can do the work in } &\frac{30}{5} = 6 \text{ days}\end{aligned}$$

Hence, A and B can do the work in 6 days.

Question 3

A and B together can do a piece of work in $6\frac{2}{3}$ days, but B alone can do it in 10 days. How long will A take to do it alone?

Answer

A + B can do the work in $6\frac{2}{3} = \frac{20}{3}$ days.

$$(A + B)\text{'s 1 day work} = \frac{3}{20}$$

$$B\text{'s 1 day work} = \frac{1}{10}$$

$$A\text{'s 1 day work} = \frac{3}{20} - \frac{1}{10}$$

$$= \frac{3 - 2}{20}$$

$$= \frac{1}{20}$$

$$A \text{ can do the work in } \frac{20}{1} = 20 \text{ days}$$

Hence, A can do the work in 20 days.

Question 4

A can do a work in 15 days and B in 20 days. If they work together on it for 4 days, what fraction of the work will be left?

Answer

$$\text{A's 1 day work} = \frac{1}{15}$$

$$\text{B's 1 day work} = \frac{1}{20}$$

$$(\text{A} + \text{B})\text{'s 1 day work} = \frac{1}{15} + \frac{1}{20}$$

$$= \frac{4 + 3}{60}$$

$$= \frac{7}{60}$$

$$(\text{A} + \text{B})\text{'s 4 day work} = \frac{7}{60} \times 4$$

$$= \frac{7}{15}$$

$$\text{Remaining work} = 1 - \frac{7}{15}$$

$$= \frac{15}{15} - \frac{7}{15}$$

$$= \frac{8}{15}$$

Hence, $\frac{8}{15}$ work will be left.

Question 5

A, B and C can do a piece of work in 6 days, 12 days and 24 days respectively.

In what time will they altogether do it?

Answer

$$\text{A's 1 day work} = \frac{1}{6}$$

$$\text{B's 1 day work} = \frac{1}{12}$$

$$\text{C's 1 day work} = \frac{1}{24}$$

$$(\text{A} + \text{B} + \text{C})\text{'s 1 day work} = \frac{1}{6} + \frac{1}{12} + \frac{1}{24}$$

$$= \frac{4 + 2 + 1}{24}$$

$$= \frac{7}{24}$$

$$\text{A} + \text{B} + \text{C} \text{ can do the work in } \frac{24}{7} = 3\frac{3}{7} \text{ days}$$

Hence, A + B + C can do the work in $3\frac{3}{7}$ days.

Question 6

A and B working together can mow a field in 56 days and with the help of C, they could have mowed it in 42 days. How long would C take to mow the field by himself?

Answer

$$(\text{A} + \text{B})\text{'s 1 day work} = \frac{1}{56}$$

$$(\text{A} + \text{B} + \text{C})\text{'s 1 day work} = \frac{1}{42}$$

$$\text{C's 1 day work} = \frac{1}{42} - \frac{1}{56}$$

$$= \frac{4 - 3}{168}$$

$$= \frac{1}{168}$$

C can do the work in $\frac{168}{1}$ days

Hence, C can do the work in 168 days.

Question 7

A can do a piece of work in 24 days, A and B can do it in 16 days and A, B and C in $10\frac{2}{3}$ days. In how many days can A and C do it working together?

Answer

$$\text{A's 1 day work} = \frac{1}{24}$$

$$(\text{A} + \text{B})\text{'s 1 day work} = \frac{1}{16}$$

$$\text{B's 1 day work} = (\text{A} + \text{B})\text{'s 1 day work} - \text{A's 1 day work}$$

$$= \frac{1}{16} - \frac{1}{24}$$

$$= \frac{(3 - 2)}{48}$$

$$= \frac{1}{48}$$

$$(\text{A} + \text{B} + \text{C})\text{'s 1 day work} = \frac{3}{32}$$

$$(\text{A} + \text{C})\text{'s 1 day work} = (\text{A} + \text{B} + \text{C})\text{'s 1 day work} - \text{B's 1 day work}$$

$$= \frac{3}{32} - \frac{1}{48}$$

$$= \frac{9 - 2}{96}$$

$$= \frac{7}{96}$$

A + C can do the work in $\frac{96}{7} = 13\frac{5}{7}$ days

Hence, A + C can do the work in $13\frac{5}{7}$ days.

Question 8

A can do a piece of work in 20 days and B in 15 days. They worked together on it for 6 days and then A left. How long will B take to finish the remaining work?

Answer

$$\text{A's 1 day work} = \frac{1}{20}$$

$$\text{B's 1 day work} = \frac{1}{15}$$

$$(\text{A} + \text{B})\text{'s 1 day work} = \frac{1}{20} + \frac{1}{15}$$

$$= \frac{3 + 4}{60}$$

$$= \frac{7}{60}$$

$$(\text{A} + \text{B})\text{'s 6 day work} = \frac{7}{60} \times 6$$

$$= \frac{7}{10}$$

$$\text{Remaining work} = 1 - \frac{7}{10}$$

$$= \frac{10}{10} - \frac{7}{10}$$

$$= \frac{3}{10}$$

$$\text{No. of days taken by B to finish the remaining work} = \frac{\text{Remaining work}}{\text{B's 1 day work}} =$$

$$\frac{\frac{3}{10}}{\frac{1}{15}}$$

$$= \frac{3 \times 15}{10 \times 1}$$

$$= \frac{45}{10}$$

$$= \frac{9}{2}$$

$$= 4\frac{1}{2}$$

Hence, B will take $4\frac{1}{2}$ days to finish the remaining work.

Question 9

A can finish a piece of work in 15 days and B can do it in 10 days. They worked together for 2 days and then B goes away. In how many days will A finish the remaining work?

Answer

$$\text{A's 1 day work} = \frac{1}{15}$$

$$\text{B's 1 day work} = \frac{1}{10}$$

$$(\text{A} + \text{B})\text{'s 1 day work} = \frac{1}{15} + \frac{1}{10}$$

$$= \frac{2 + 3}{30}$$

$$= \frac{5}{30}$$

$$= \frac{1}{6}$$

$$(\text{A} + \text{B})\text{'s 2 day work} = \frac{1}{6} \times 2$$

$$= \frac{1}{3}$$

$$\text{Remaining work} = 1 - \frac{1}{3}$$

$$= \frac{3}{3} - \frac{1}{3}$$

$$= \frac{2}{3}$$

$$\text{No. of days taken by A to finish the remaining work} = \frac{\text{Remaining work}}{\text{B's 1 day work}} =$$

$$\frac{\frac{2}{3}}{\frac{1}{15}}$$

$$\begin{aligned} &= \frac{2 \times 15}{3 \times 1} \\ &= \frac{30}{3} \\ &= 10 \end{aligned}$$

Hence, A will take 10 days to finish the remaining work.

Question 10

A can do a piece of work in 10 days, B in 18 days, and A, B and C together in 4 days. In what time would C do it alone?

Answer

$$\text{A's 1 day work} = \frac{1}{10}$$

$$\text{B's 1 day work} = \frac{1}{18}$$

$$(\text{A} + \text{B} + \text{C})\text{'s 1 day work} = \frac{1}{4}$$

$$\text{C's 1 day work} = \frac{1}{4} - \left(\frac{1}{10} + \frac{1}{18} \right)$$

$$= \frac{1}{4} - \left(\frac{9 + 5}{90} \right)$$

$$= \frac{1}{4} - \frac{14}{90}$$

$$= \frac{1}{4} - \frac{7}{45}$$

$$= \frac{45 - 28}{180}$$

$$= \frac{17}{180}$$

$$\text{C can do the work in } \frac{180}{17} = 10\frac{10}{17}$$

Hence, C can do the work alone in $10\frac{10}{17}$ days.

Question 11

A can do $\frac{1}{4}$ of a work in 5 days and B can do $\frac{1}{3}$ of the same work in 10 days.

Find the number of days in which both working together will complete the work.

Answer

$$\text{A's 5 days work} = \frac{1}{4}$$

$$\text{A's 1 day work} = \frac{1}{4 \times 5}$$

$$= \frac{1}{20}$$

$$\text{B's 10 days work} = \frac{1}{3}$$

$$\text{B's 1 day work} = \frac{1}{3 \times 10}$$

$$= \frac{1}{30}$$

$$(\text{A} + \text{B})\text{'s 1 day work} = \frac{1}{20} + \frac{1}{30}$$

$$= \frac{(3 + 2)}{60}$$

$$= \frac{5}{60}$$

$$= \frac{1}{12}$$

A and B can do the work in $\frac{12}{1}$ days

Hence, A and B working together can complete the work in 12 days.

Question 12

One tap can fill a cistern in 3 hours and the waste pipe can empty the full cistern in 5 hours. In what time will the empty cistern be full, if the tap and the waste pipe are kept open together?

Answer

$$\text{Tap's 1 hour work} = \frac{1}{3}$$

$$\text{Waste tap's 1 hour work} = -\frac{1}{5}$$

$$\text{Both tap's 1 hour work} = \frac{1}{3} - \frac{1}{5}$$

$$= \frac{5 - 3}{15}$$

$$= \frac{2}{15}$$

$$\text{Total time taken by both pipe} = \frac{15}{2} \text{ hours}$$

Hence, with both taps kept open, the empty cistern will be full in $7\frac{1}{2}$ hours.

Test Yourself

Question 1(i)

The value of a when

x	y
2	6
a	18
9	27

is:

1. 1
2. 1.5
3. 9
4. 6

Answer

$$\text{As } \frac{2}{6} = \frac{1}{3}$$

$$\text{And } \frac{9}{27} = \frac{1}{3}$$

So, we can say x and y vary directly

$$\frac{2}{6} = \frac{a}{18}$$

$$\Rightarrow a = \frac{2 \times 18}{6}$$

$$\Rightarrow a = \frac{36}{6}$$

$$\Rightarrow a = 6$$

Hence, option 4 is the correct option.

Question 1(ii)

The value of b when

x	y
3	12
b	9
12	3

is:

1. 4
2. 36
3. 24
4. none of these

Answer

$$3 \times 12 = 36$$

$$12 \times 3 = 36$$

So we can say x and y vary inversely

$$a_1 b_1 = a_2 b_2 \quad [\because a \text{ varies inversely to } b]$$

$$\Rightarrow 3 \times 12 = b \times 9$$

$$\Rightarrow 36 = b \times 9$$

$$\Rightarrow b = \frac{36}{9}$$

$$\Rightarrow b = 4$$

Hence, option 1 is the correct option.

Question 1(iii)

15 note books can be bought for ₹ 240. The number of note books that can be bought for ₹ 160 is:

1. 16

2. 10

3. 18

4. 15

Answer

Let x notebooks cost ₹ 160.

Quantity	Cost
15	₹ 240
x	₹ 160

Since it is a case of direct variation,

$$\therefore \frac{15}{240} = \frac{x}{160}$$

$$\Rightarrow x = \frac{15 \times 160}{240}$$

$$\Rightarrow x = \frac{2,400}{240}$$

$$\Rightarrow x = 10$$

Hence, option 2 is the correct option.

Question 1(iv)

6 men can do a certain piece of work in 15 days. The number of men required to complete the same work in 10 days:

1. 30

2. 25

3. 60

4. 9

Answer

Let x men be required to complete the work in 10 days.

Men	Days
6	15
x	10

Since it is a case of inverse variation,

$$\therefore 6 \times 15 = x \times 10$$

$$\Rightarrow 90 = 10x$$

$$\Rightarrow x = \frac{90}{10}$$

$$\Rightarrow x = 9$$

Hence, option 4 is the correct option.

Question 1(v)

If x is in inverse variation with y and x = 4 when y = 6, the value of x when y = 12 is:

1. 2

2. 18

3. 8

4. 12

Answer

Since it is a case of inverse variation,

$$\therefore 4 \times 6 = x \times 12$$

$$\Rightarrow 24 = 12x$$

$$\Rightarrow x = \frac{24}{12}$$

$$\Rightarrow x = 2$$

Hence, option 1 is the correct option.

Question 1(vi)

Statement 1: Work done varies inversely to the number of persons at work.

Statement 2: Time taken to finish a work varies directly to the number of persons at work.

Which of the following options is correct?

1. Both the statements are true.

2. Both the statements are false.

3. Statement 1 is true, and statement 2 is false.

4. Statement 1 is false, and statement 2 is true.

Answer

Since, Work done is constant.

Work done increases with more people working. Therefore, work done is directly proportional to the number of persons.

So, statement 1 is false.

Time taken decreases with more people working.

Therefore, time taken is inversely proportional to the number of persons.

So, statement 2 is false.

Hence, option 2 is the correct option.

Question 1(vii)

Assertion (A) : In the following table, p and q are in direct variation.

p	3	2	4
q	8	12	6

Reason (R) : If two quantities p and q are in direct variation, then $\frac{p}{q}$ is always constant.

1. Both A and R are correct, and R is the correct explanation for A.

2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

If two numbers p and q are in direct variation, then $\frac{p}{q}$ will be constant.

So, reason is true.

For table,

$$\Rightarrow \frac{p_1}{q_1} = \frac{3}{8}$$

$$\Rightarrow \frac{p_2}{q_2} = \frac{2}{12} = \frac{1}{6}$$

$$\Rightarrow \frac{p_3}{q_3} = \frac{4}{6} = \frac{2}{3}$$

$$\therefore \frac{p_1}{q_1} \neq \frac{p_2}{q_2} \neq \frac{p_3}{q_3}$$

So, assertion (A) is false.

Hence, option 4 is the correct option.

Question 1(viii)

Assertion (A) : If 3 pipes can fill a tank in 6 hours, then 6 pipes will fill the same tank in half the time.

Reason (R) : Time required to complete a certain work =

$$\frac{\text{Work to be completed}}{\text{One day's work}}.$$

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

If 3 pipes take 6 hours, then their combined rate of filling the tank is $\frac{1}{6}$ tank/hr.

Each pipe's rate = $\frac{1}{6} \times \frac{1}{3} = \frac{1}{18}$ tank/hr

Thus, 6 pipes will fill = $6 \times \frac{1}{18} = \frac{1}{3}$ tank in an hr.

Time to fill 1 tank by 6 pipes = 3 hours

So, assertion (A) is true.

Time required to complete a certain work = $\frac{\text{Work to be completed}}{\text{One day's work}}.$

This is a standard formula used in time and work problems.

So, reason (R) is true but reason (R) does not clearly explains assertion (A).

Hence, option 2 is the correct option.

Question 1(ix)

Assertion (A) : In the following table, p and q are in inverse variation.

p	5	2	4
q	10	25	12.5

Reason (R) : If two quantities p and q are in inverse variation, then $\frac{p}{q}$ is always constant.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

We know that,

For two numbers to be in inverse variation,

$$p \times q = \text{constant}$$

So, reason (R) is false.

From table,

$$\Rightarrow x_1 y_1 = 5 \times 10 = 50,$$

$$\Rightarrow x_2 y_2 = 2 \times 25 = 50,$$

$$\Rightarrow x_3 y_3 = 4 \times 12.5 = 50.$$

So, assertion (A) is true.

Hence, option 3 is the correct option.

Question 1(x)

Assertion (A) : The cost of 16 bulbs is ₹ 144. The number of bulbs than can be bought for ₹ 270 is 30.

Reason (R) : In inverse variation, the ratio of one kind of like terms is equal to the inverse ratio of the second kind of like terms.

1. Both A and R are correct, and R is the correct explanation for A.
2. Both A and R are correct, and R is not the correct explanation for A.
3. A is true, but R is false.
4. A is false, but R is true.

Answer

Given,

The cost of 16 bulbs = ₹ 144

$$\text{The cost of 1 bulb} = \frac{144}{16} = ₹ 9$$

$$\text{Number of bulbs for ₹ 270} = \frac{\text{Total money}}{\text{Price of each bulb}} = \frac{270}{9} = 30$$

So, assertion (A) is true.

In inverse variation,

$$\Rightarrow x_1 y_1 = x_2 y_2$$

$$\Rightarrow \frac{x_1}{x_2} = \frac{y_2}{y_1}$$

So, reason (R) is true but, reason (R) does not explain assertion (A).

Hence, option 2 is the correct option.

Question 2

If x varies directly as y and x = 150 when y = 50. Find:

1. x, when y = 12.5

2. y, when x = 75

Answer

1. Since it is a case of direct variation,

$$\frac{150}{50} = \frac{x}{12.5}$$

$$\Rightarrow \frac{3}{1} = \frac{x}{12.5}$$

$$\Rightarrow x = 3 \times 12.5$$

$$\Rightarrow x = 37.5$$

Hence, when $y = 12.5$, then $x = 37.5$.

2. Since it is a case of direct variation,

$$\frac{150}{50} = \frac{75}{y}$$

$$\Rightarrow \frac{3}{1} = \frac{75}{y}$$

$$\Rightarrow y = \frac{75}{3}$$

$$\Rightarrow x = 25$$

Hence, when $x = 75$, then $y = 25$.

Question 3

If x varies inversely as y and $y = 300$ when $x = 60$. Find:

1. x , when $y = 90$

2. y , when $x = 300$

Answer

1. Since it is a case of inverse variation,

$$60 \times 300 = x \times 90$$

$$\Rightarrow 18000 = x \times 90$$

$$\Rightarrow x = \frac{18000}{90}$$

$$\Rightarrow x = 200$$

Hence, when $y = 90$, then $x = 200$.

2. Since it is a case of inverse variation,

$$60 \times 300 = 300 \times y$$

$$\Rightarrow 18000 = 300 \times y$$

$$\Rightarrow y = \frac{18000}{300}$$

$$\Rightarrow y = 60$$

Hence, when $x = 300$, then $y = 60$.

Question 4

Total length of 153 iron bars is 680 m. What will be the total length of 135 similar bars.

Answer

Let x m be total length of 135 bars.

Iron bars	Length (in m)
153	680
135	x

Since it is a case of direct variation,

$$\therefore \frac{153}{680} = \frac{135}{x}$$

$$\Rightarrow x = \frac{135 \times 680}{153}$$

$$\Rightarrow x = \frac{91,800}{153}$$

$$\Rightarrow x = 600$$

Hence, 600 m is the total length of 135 bars.

Question 5

12 men can repair a road in 25 days; how long will 30 men will take to do so?

Answer

Let 30 men take x days to repair the road.

Men	Days
12	25
30	x

Since it is a case of inverse variation,

$$\therefore 12 \times 25 = 30 \times x$$

$$\Rightarrow 300 = 30x$$

$$\Rightarrow x = \frac{300}{30}$$

$$\Rightarrow x = 10$$

Hence, 30 men will do the work in 10 days.

Question 6

The price of oranges is ₹ 90 per dozen. Manoj can buy 12 dozen oranges with the money he has. If the price of orange is increased by ₹ 30, how many oranges can Manoj buy?

Answer

Cost of 1 dozen oranges = ₹ 90

Cost of 12 dozen oranges = ₹ 90 x 12 = ₹ 1,080

Price of oranges is increased by ₹ 30,

$\therefore$ New price of 1 dozen oranges = ₹ (90 + 30) = ₹ 120

Let Manoj buy x dozen oranges for ₹ 120 per dozen.

Oranges (in dozen)	Cost
12	90

Oranges (in dozen)	Cost
x	120

As the price of oranges has increased, so now Manoj will be able to buy lesser number of oranges with the money he has.

∴ It is a case of inverse variation.

$$\therefore 12 \times 90 = x \times 120$$

$$\Rightarrow 1080 = x \times 120$$

$$\Rightarrow x = \frac{1080}{120}$$

$$\Rightarrow x = 9$$

Hence, Manoj can buy 9 dozen oranges for ₹ 1,080.

Question 7

A and B can do a work in 8 days, B and C in 12 days, and A and C in 16 days. In what time can they do it, all working together?

Answer

$$(A + B)\text{'s 1 day work} = \frac{1}{8}$$

$$(B + C)\text{'s 1 day work} = \frac{1}{12}$$

$$(C + A)\text{'s 1 day work} = \frac{1}{16}$$

$$2(A + B + C)\text{'s 1 day work} = \frac{1}{8} + \frac{1}{12} + \frac{1}{16}$$

$$= \frac{6 + 4 + 3}{48}$$

$$= \frac{13}{48}$$

$$(A + B + C)\text{'s 1 day work} = \frac{13}{48 \times 2}$$

$$= \frac{13}{96}$$

No. of days to complete the work when A, B and C are working together =

$$\frac{96}{13} = 7\frac{5}{13}$$

Hence, A, B and C working together can complete the work in $7\frac{5}{13}$ days.

Question 8

A and B complete a piece of work in 24 days. B and C do the same work in 36 days, and A, B and C together finish it in 18 days. In how many days will:

(i) A alone,

(ii) C alone,

(iii) A and C together, complete the work?

Answer

$$(i) (A + B)'s \text{ 1 day work} = \frac{1}{24}$$

$$(B + C)'s \text{ 1 day work} = \frac{1}{36}$$

$$(A + B + C)'s \text{ 1 day work} = \frac{1}{18}$$

$$A's \text{ 1 day work} = (A + B + C)'s \text{ 1 day work} - (B + C)'s \text{ 1 day work}$$

$$= \frac{1}{18} - \frac{1}{36}$$

$$= \frac{2 - 1}{36}$$

$$= \frac{1}{36}$$

Number of days required by A alone = 36 days

Hence, A requires 36 days to complete the work alone.

$$(ii) C's \text{ 1 day work} = (A + B + C)'s \text{ 1 day work} - (A + B)'s \text{ 1 day work}$$

$$= \frac{1}{18} - \frac{1}{24}$$

$$= \frac{4 - 3}{72}$$

$$= \frac{1}{72}$$

Number of days required by C alone = 72 days

Hence, C requires 72 days to complete the work alone.

$$\begin{aligned}\text{(iii) (A + C)'s 1 day work} &= \frac{1}{36} + \frac{1}{72} \\ &= \frac{2 + 1}{72} \\ &= \frac{3}{72}\end{aligned}$$

No. of days required to complete the work with A and C working together =
 $\frac{72}{3} = 24$ days

Hence, A and C together will complete the work in 24 days.

Question 9

A and B can do a piece of work in 40 days, B and C in 30 days, and C and A in 24 days.

(i) How long will it take them to do the work, working together?

(ii) In what time can each finish it working alone?

Answer

$$(A + B)\text{'s 1 day work} = \frac{1}{40}$$

$$(B + C)\text{'s 1 day work} = \frac{1}{30}$$

$$(C + A)\text{'s 1 day work} = \frac{1}{24}$$

$$2(A + B + C)\text{'s 1 day work} = \frac{1}{40} + \frac{1}{30} + \frac{1}{24}$$

$$= \frac{(3 + 4 + 5)}{120}$$

$$= \frac{12}{120}$$

$$= \frac{1}{10}$$

$$(A + B + C)'s\ 1\ day\ work = \frac{1}{2 \times 10}$$

$$= \frac{1}{20}$$

No. of days required to complete the work when A, B and C are working together = 20 days

Hence, A, B and C working together can complete the work in 20 days.

(ii) A's 1 day work = (A + B + C)'s 1 day work - (B + C)'s 1 day work

$$= \frac{1}{20} - \frac{1}{30}$$

$$= \frac{(3 - 2)}{60}$$

$$= \frac{1}{60}$$

∴ A alone will complete the work in 60 days.

B's 1 day work = (A + B + C)'s 1 day work - (C + A)'s 1 day work

$$= \frac{1}{20} - \frac{1}{24}$$

$$= \frac{(6 - 5)}{120}$$
$$= \frac{1}{120}$$

∴ **B alone will complete the work in 120 days.**

C's 1 day work = (A + B + C)'s 1 day work - (A + B)'s 1 day work

$$= \frac{1}{20} - \frac{1}{40}$$
$$= \frac{(2 - 1)}{40}$$
$$= \frac{1}{40}$$

∴ **C alone will complete the work in 40 days.**

Question 10

A can do a piece of work in 10 days, B in 12 days and C in 15 days. All begin together but A leaves the work after 2 days and B leaves 3 days before the work is finished. How long did the work last?

Answer

$$\text{A's 1 day work} = \frac{1}{10}$$

$$\text{B's 1 day work} = \frac{1}{12}$$

$$\text{C's 1 day work} = \frac{1}{15}$$

Let the work get completed in x days.

$\therefore$ A's 2 days work + B's $(x - 3)$ days work + C's x days work = 1

$$\Rightarrow \frac{1}{10} \times 2 + \frac{1}{12} \times (x - 3) + \frac{1}{15} \times x = 1$$

$$\Rightarrow \frac{1}{5} + \frac{(x - 3)}{12} + \frac{x}{15} = 1$$

$$\Rightarrow \frac{[12 + 5(x - 3) + 4x]}{60} = 1$$

$$\Rightarrow \frac{(12 + 5x - 15 + 4x)}{60} = 1$$

$$\Rightarrow \frac{(9x - 3)}{60} = 1$$

$$\Rightarrow 9x - 3 = 1 \times 60$$

$$\Rightarrow 9x - 3 = 60$$

$$\Rightarrow 9x = 60 + 3$$

$$\Rightarrow 9x = 63$$

$$\Rightarrow x = \frac{63}{9}$$

$$\Rightarrow x = 7$$

Hence, the work lasts for 7 days.

Question 11

Two pipes P and Q would fill an empty cistern in 24 minutes and 32 minutes respectively. Both the pipes being opened together, find when the first pipe must be turned off so that the empty cistern may be just filled in 16 minutes.

Answer

$$\text{Work done by P in 1 minute} = \frac{1}{24}$$

$$\text{Work done by Q in 1 minute} = \frac{1}{32}$$

Let the first pipe be turned off after x minutes.

Then, P's x minutes work + Q's 16 minutes work = 1

$$\Rightarrow \frac{1}{24} \times x + \frac{1}{32} \times 16 = 1$$

$$\Rightarrow \frac{x}{24} + \frac{1}{2} = 1$$

$$\Rightarrow \frac{x}{24} = 1 - \frac{1}{2}$$

$$\Rightarrow \frac{x}{24} = \frac{1}{2}$$

$$\Rightarrow x = \frac{1}{2} \times 24$$

$$\Rightarrow x = 12$$

Hence, pipe P must be closed after 12 minutes.

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